

QP Code : **NP-19836**

(3 Hours)

[Total Marks : 80]

N.B. : (1) Questions No.1 is **compulsory**.(2) Attempt any **three** questions out of remaining **five** questions.(3) Assumptions made should be **clearly** stated.(4) **Figures** to the **right** indicate **full** marks(5) Assume **suitable** data wherever **required** but **justify** the same.

1. (a) Differentiate between NFA and DFA. 5
- (b) Explain CNF and GNF with example. 5
- (c) State and prove closure properties of Context Free Languages. 5
- (d) Give Applications of Regular Expression and Finite Automata. 5

2. (a) Construct an NFA with epsilon transition for following RE. 5
 $(00 + 11)^*(10)^*$
- (b) Give formal definition of Regular expression. Give R.E. for following :— 5
 - (i) Set of all strings over $\{1, 0\}$ that end with 1 and has no substring 00.
 - (ii) Set of all strings over $\{1, 0\}$ with even number of 1's followed by odd number of 0's.
- (c) Compare and Contrast Moore and Mealy Machine. Construct Moore Machine 10
 to find out the residue-modulo-3 for binary numbers.

3. (a) Consider the following grammar :— 10

$$S \rightarrow i C t S \mid i C t S e S \mid a$$

$$C \rightarrow b$$
 For the String 'ibtibtaea' find the following :
 - (i) Leftmost derivation
 - (ii) Rightmost derivation
 - (iii) Parse Tree
 - (iv) Check if the above grammar is Ambiguous
- (b) Design PDA that checks for well- formed parentheses. 10

4. (a) Design a TM that recognizes palindrome strings where $\Sigma = \{0, 1\}$ 10
- (b) Construct NFA that accepts a set of all strings over $\{a, b\}$ ending with "abb" Convert this NFA to Equivalent DFA. 10

[TURN OVER]

5. (a) Convert the following Grammar to CNF form :—

10

$$S \rightarrow ABA$$

$$A \rightarrow aA \mid bA \mid \epsilon$$

$$B \rightarrow bB \mid aA \mid \epsilon$$

(b) Give and explain the formal statement of Pumping Lemma for regular languages and use it to prove that the following language is not regular :

$$L = \{ a^n b^n \mid n \geq 1 \}$$

6. Write short note on :—

20

(a) Chomsky Hierarchy of Grammar

(b) Variants of Turing Machine

(c) Rice's Theorem

(d) Recursive and Recursively enumerable languages.